3.1 Operations Management

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Operations management

- Operations management is a field of business concerned with the administration of business practices to maximize efficiency within an organization.

- It involves planning, organizing, and overseeing the organization’s processes to balance revenues and costs and achieve the highest possible operating profit.

- An operations manager is tasked with ensuring that the organization successfully converts inputs such as materials, labor, and technology into outputs in an efficient manner.

Operations management

- Operations management involves managing the operations and processes of an organization.

- Some of the functions performed by an operations manager include supply chain management, product design, forecasting, quality control, and delivery management.

- An operations manager must have excellent organizational, coordination, and people skills and must be tech-savvy.

Responsibilities in Operations Management

- Operations management is a field of business that involves managing the operations of a business to ensure efficiency in the execution of projects.

- It means that the individual in charge of the department will be required to perform various strategic functions. Some of the functions include:

Functions of Operations management

Functions of Operations management

1. Product Design

- Product design involves creating a product that will be sold to the end consumer. It involves generating new ideas or expanding on current ideas in a process that will lead to the production of new products. The operations manager's responsibility is to ensure that the products sold to consumers meet their needs, as well as match current market trends.

- Consumers are more interested in the quality of the product more than the quantity, and the organization should create systems that ensure the products produced meet the needs of the consumer.

Functions of Operations management

2. Forecasting

- Forecasting involving making predictions of events that will occur in the future based on past data. One of the events that the operations manager is required to predict is the consumer demand for the company's products.

- The manager relies on past and present data on the uptake of the company's products to determine future trends in consumption. The forecasts help the company know the volume of products needed to meet the market demand.

Functions of Operations management

3. Supply Chain Management

- Supply chain management involves managing the production process from raw materials to the finished product. It controls everything from production, shipping, distribution, to delivery of products.

- The operations manager manages the supply chain process by maintaining control of inventory management, the production process, distribution, sales, and sourcing of suppliers to supply required goods at reasonable prices. A properly managed supply chain process will result in an efficient production process, low overhead costs, and timely delivery of products to consumers.

Functions of Operations management

4. Delivery Management

- The operations manager is in charge of delivery management. The manager ensures that the goods are delivered to the consumer in a timely manner. They must follow up with consumers to ensure that the goods delivered are what the consumers ordered and that they meet their functionality needs.

- If the customer is unsatisfied with the product or is complaining about certain features of the product, the operations manager receives the feedback and forwards it to the relevant departments.

Skills of an Operations Manager

- Operations management is a cross-department role where the manager assumes an array of responsibilities across multiple disciplines. To be successful, an operations manager must possess the following skills:

1. Organizational Abilities

- Organizational abilities refer to the ability of the operations manager to focus on different projects without getting distracted by the many processes. The operations manager should be able to plan, execute, and monitor each project to the end without losing focus.

- If a manager is not organized, uncompleted tasks will pile up, important documents will get lost in the process, and a majority of the time will be spent finding lost documents that could be easily accessible had the manager been organized. Good organization skills can increase production efficiency and help the manager save time.

Skills of an Operations Manager

2. Coordination

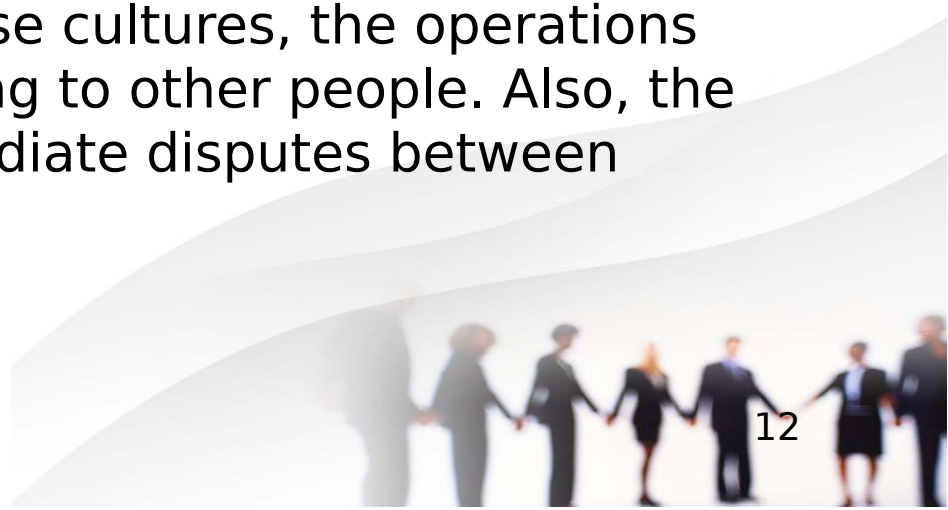
- An operations manager needs to have good coordination by knowing how to integrate resources, activities, and time to ensure proper use of the resources toward the achievement of the organization’s goals. Coordination involves carrying out specific activities simultaneously and switching between the activities with ease. It also involves dealing with interruptions, obstacles, and crises, and efficiently going back to the normal routine functions to prevent further interruptions.

Skills of an Operations Manager

3. People Skills

- Most of the responsibilities of an operations manager involve dealing with people. This means that they must know how to relate with the employees, outside stakeholders, and other members of senior management. An operations manager should know how to manage the fine lines with other colleagues by knowing how to communicate, listen, and relate to them on professional and personal levels.

- Since workplaces are made up of people from diverse cultures, the operations manager needs to show tolerance and understanding to other people. Also, the manager should be able to resolve conflicts and mediate disputes between employees and members of the senior staff.



Skills of an Operations Manager

4. Tech-savvy

- In this age of rapidly advancing technologies, an operations manager needs to have an affinity for technology in order to be in a position to design processes that are both efficient and tech-compliant. Modern organizations are becoming increasingly tech-dependent in order to gain a competitive advantage in the market.

- This means that most of the processes conducted manually, such as procurement, must transition to more efficient automated processes. When an operations manager is familiar with the latest innovations in the tech industry, they can use the innovations to improve internal processes.

Foundations of OM:

- In its broadest sense, operations management is responsible for all aspects of creating goods and services. It manages resources such as materials, machines, technology, and people, and makes products and services that the marketplace wants. The whole chain of events must be well managed for a business to be competitive.

- As we define operations management more fully, we consider these
- Planning: Operations managers must constantly forecast, plan, and adjust to optimize processes based on conditions.
- Process: Production of goods or services requires having strong, repeatable processes.
- Efficiency: Managers must troubleshoot bottlenecks, inadequate resources, and downtimes to create optimal efficiency.
- Cost Control: Production is typically a major part of a company's cost structure, and you must manage it wisely.
- Quality: Good quality control is necessary to maintain customer satisfaction and the company's reputation. Companies can greatly suffer without it.
- Continuous Improvement: To remain competitive, companies need to have processes in place to consistently seek better ways of doing things.
- Technology: Underlying all of these foundations is technology. Well-used technology keeps a company ahead of the curve.
- Profitability: Executed properly, all of the above foundations lead to a strong bottom line.

The Importance of Operations Management

- Operations management serves as the engine room of the organization, planning and driving manufacturing or services. To compete in an ever-changing market, operations managers must maximize efficiency, productivity, and profit, which have always been vital to a company's survival. At most companies, there are many operations employees, and the department's budget is large.

- We can also see the importance of operations management in these aspects of a company's success:

- Customer service
- Product or service quality
- Correctly-functioning processes
- Market competitiveness
- Technological advances
- Profitability

Functions and Roles in Operations Management

The field of operations management includes a diverse set of functions and roles, which can differ among industries and different-sized companies:

- Planning and implementing manufacturing plants
- Managing projects
- Planning information systems
- Helping to design and develop products and services
- Managing inventory through the supply chain
- Managing delivery to customers in a timely manner
- Optimizing quality control
- Conducting procurement/purchasing
- Managing logistics
- Managing transportation and distribution
- Managing and maintaining facilities
- Conducting enterprise resource planning (ERP)
- Forecasting for planning
- Planning for capacity
- Navigating industrial labor relations
- Analyzing the value chain
- Optimizing resource usage
- Eliminating waste and bottlenecks
- Continuously improving processes
- Executing a company’s strategic plan

Operations Management Strategies

- Use of Data: Analytics are essential for strong planning, adjustments, and decision making. Two common types are efficiency metrics and effectiveness metrics.
- Inventory Analysis: To manage inventory in the supply chain, ABC analysis (also called Pareto analysis) comes into play. It divides inventory into three categories A, B, and C; “A” has the most value and tightest controls, and “C” has the least.
- Data Challenges: Data is often siloed, which makes it difficult to compare. But newer systems and setups are making this easier and helping analysts and managers examine data in new, helpful ways.
- Process Design: Researching, forecasting, and developing a sound process takes expertise and energy, but the results can be lasting.
- Forecasting and Goal Setting: The best forecasting often combines a look at historical data with analysis of changing conditions.
- Collaboration Among Departments: With good communication and collaboration, operations management can work effectively with finance, sales, marketing, human resources, and other departments.
- Being Green: Ecological soundness has become a strategic and legal necessity at companies nowadays, especially in manufacturing.
- Managing People: With all the advancements in machinery and technology, people remain critical to the equation, though often in different types of jobs.

Levels of Operations Management

Three Levels of Operations Management

- Strategic,
- Tactical, and
- Operations.

- To achieve the company's goals, operations managers develop strategies.
- Under those broad strategies are tactics, or specific tasks and steps to implement the strategies.
- The third level, operations, refers to the daily control of production, such as scheduling, monitoring, and adjusting.

Principles of Operations Management

- Randall Schaefer, CPIM, described The 10 Principles of Operations Management

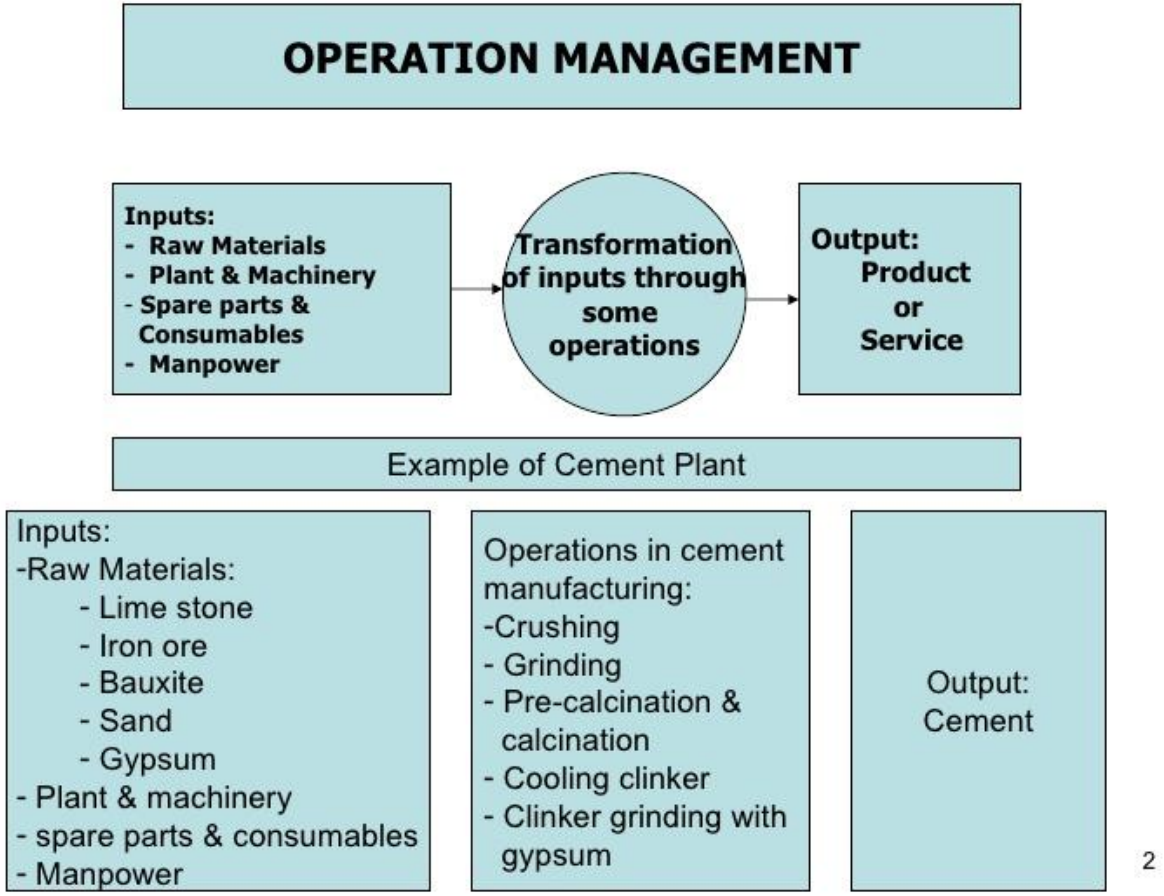
- Reality: There is no universal solution to the problems in your business.
- Organization: You must organize all aspects of production into a coherent whole.
- Fundamentals: Adhere to fundamentals, such as accurate inventory records.
- Accountability: People try harder when they're held accountable.
- Variance: Variance is part of every process.
- Causality: Problems are often symptoms. Get to the root cause.
- Managed Passion: People with a passion for their jobs will drive your company.
- Humility: You don't have to know everything.
- Success: Define success, and change with the market.

Importance value of Operations Management

Importance and Value of Operations Management

1. Production Efficiency
2. Profit Maximization
3. Satisfaction of the Customers
4. Effective Utilization of the Resources
5. Management of Inventory
6. Capacity Management
7. Improvement of Product Quality
8. Create Strong Operation Skill
9. Improvement of Working Conditions
10. Supply Chain Management
11. Scheduling Management

Operations management in Cement Plant



Current Trends in Operations Management

- Business process reengineering, or BPR, which helps companies revamp their organizations from the ground up
- Lean manufacturing, Six Sigma, and Agile are disciplines focused on efficient, adaptable production and continue to be mainstay approaches
- Reconfigurable manufacturing systems, designed for flexibility with sudden market changes
- Behavioral operations management, which focuses on human behavior as it relates to operations management
- Sustainability, or maintaining ecologically minded practices, under changing laws.

Assignment - 3.1

Q1. Explain Operations management and its functions.

Q 2. Explain the reponsibilities in operations management.

Q 3. Elaborate the skills of an Operations manager.

Q 4. Explain operations management strategies.

Q 5. Explain principle of operations management.